Math 106L In Class 2 Practice

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Position from Acceleration

1. Suppose that $s(t)$ gives the position of an object moving along a straight line in meters at time t seconds. The object starts 2 meters to the right of a reference point, moving at 3 m/s to the left. The object has a constant acceleration of 2 m/s^2. Find the velocity and position functions and describe the motion.

2. A particle starts at the reference point moving at 8 m/s to the right. It is subject to a constant acceleration of -4 m/s^2. Find $s(t)$ and describe the motion.

3. An object starts 6 meters to the left of a reference point with an initial velocity of 5 m/s to the right. It experiences a constant acceleration of -2 m/s^2. Find $s(t)$ and describe the motion.

4. Suppose an object is at $s(0) = 3$ meters and $v(0) = -4$ m/s with a constant acceleration of 2 m/s^2. Solve for $s(t)$ and describe the motion.

5. An object starts 4 meters to the right of a reference point, moving 4 m/s to the left, with a constant acceleration of 2 m/s^2. Find $s(t)$ and describe the motion.

6. A researcher on Earth ($g = 9.8$ m/s^2) stands on the edge of a 50-meter cliff and throws a stone **upward** with an initial speed of 15 m/s. Find the position function $s(t)$, the maximum height reached by the stone, and the total time it takes for the stone to hit the ground at the base of the cliff.

7. An object is thrown **downward** from the top of a 100-meter building on Earth with an initial speed of 10 m/s. Determine the position function and the velocity of the object at the moment it impacts the ground.

8. On Planet X, an astronaut **drops** a heavy tool from a height of 20 meters. The tool takes exactly 4 seconds to hit the surface. Find the value of acceleration due to gravity (g) on Planet X and write the position function $s(t)$.

9. A ball is **dropped** from a height of 30 meters on Earth. How long does it take to reach the halfway point (15 meters)?

10. On a distant moon, a rock is thrown **upward** at 4 m/s from the surface ($s(0) = 0$). It reaches a maximum height of 5 meters. Find the acceleration due to gravity (g) on this moon and determine the total time the rock is in the air.

Sums and Sigma Notation

1. Write the following sum in summation notation:

$$\frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.4} + \cdots + \frac{1}{2.4}$$

2. Write the following sum in summation notation:

$$(0.5)^2 + (1.0)^2 + (1.5)^2 + \cdots + (10)^2$$

3. Write the following sum in summation notation:

$$e^3 + e^{3.25} + e^{3.5} + e^{3.75} + \cdots + e^6$$

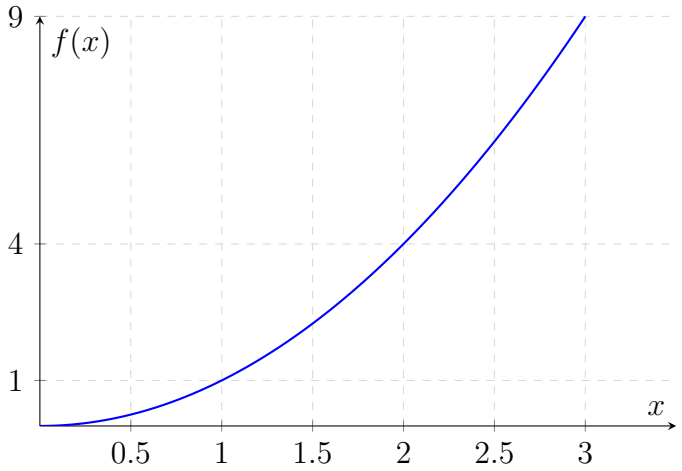
4. Write the following sum in summation notation:

$$\ln(2.1) + \ln(2.2) + \ln(2.3) + \cdots + \ln(3.5)$$

5. Write the following sum in summation notation:

$$\sqrt{5} + \sqrt{5.05} + \sqrt{5.1} + \cdots + \sqrt{6}$$

6. Consider the function $f(x) = x^2$ on the interval $[0, 2]$ with $n = 4$ subintervals.



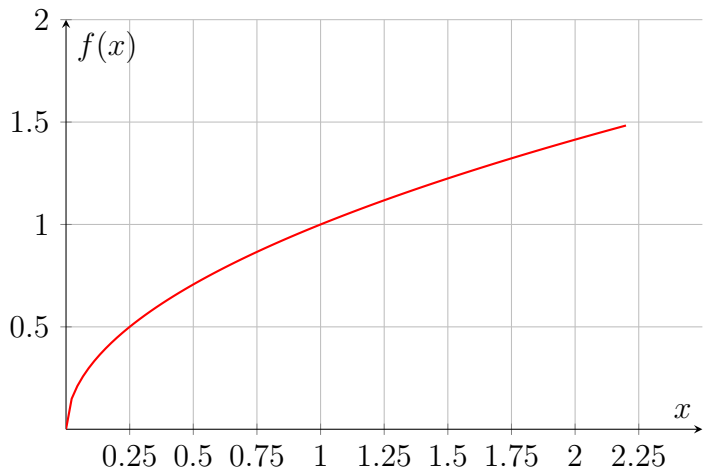
   (a) On the graph above, draw the **right-hand** rectangles used to estimate the area under the curve.

   (b) Write down the sequence of the right endpoints of the subintervals.

   (c) Write down the sequence of heights of the right-hand rectangles.

   (d) Write down the right-hand sum in sigma notation.

7. Consider the function $f(x) = \sqrt{x}$ on the interval $[0, 1.5]$ with $n = 6$ subintervals.



(a) On the graph above, draw the **left-hand** rectangles used to estimate the area under the curve.

(b) Write down the sequence of the left endpoints of the subintervals.

(c) Write down the sequence of heights of the left-hand rectangles.

(d) Write down the left-hand sum in sigma notation.

8. Consider $f(x) = \frac{1}{x}$ on the interval $[1, 3]$ with $n = 50$ subintervals.

(a) Write down the sequence of the right endpoints of the subintervals.

(b) Write down the sequence of heights of the right-hand rectangles.

(c) Write down the right-hand sum in sigma notation.

9. Consider $f(x) = \sin(x)$ on $[0, \pi]$ with $n = 100$ subintervals.

(a) Write down the sequence of the left endpoints of the subintervals.

(b) Write down the sequence of heights of the left-hand rectangles.

(c) Write down the left-hand sum in sigma notation.

10. Consider $f(x) = e^x$ on $[0, 5]$ with $n = 200$ subintervals.

(a) Write down the sequence of the right endpoints of the subintervals.

(b) Write down the sequence of heights of the right-hand rectangles.

(c) Write down the right-hand sum in sigma notation.

Left and Right Hand Sums

1. **Fill in the blanks**: A Left Hand Sum overestimates the integral of $f(x)$ when the function is _____. That is, when _____< 0. Draw a graph and rectangles on it to illustrate your answer.

2. **Fill in the blanks**: A Left Hand Sum underestimates the integral of $f(x)$ when the function is _____. That is, when _____> 0. Draw a graph and rectangles on it to illustrate your answer.

3. **Fill in the blanks**: A Right Hand Sum overestimates the integral of $f(x)$ when the function is _____. That is, when _____> 0. Draw a graph and rectangles on it to illustrate your answer.

4. **Fill in the blanks**: A Right Hand Sum underestimates the integral of $f(x)$ when the function is _____. That is, when _____< 0. Draw a graph and rectangles on it to illustrate your answer.

5. Consider $f(x) = \frac{1}{x}$ on the interval $[1, 5]$. Does a Right Hand Sum underestimate or overestimate the integral? Show any needed computations.

6. Draw a curve $f(x)$ on the interval $[1, 5]$ such that a Left Hand Sum overestimates the integral on the interval $[1, 3]$ and underestimates the integral on the interval $[3, 5]$.

   Explain what property of the function changes at $x = 3$ to cause this shift in estimation behavior. Draw rectangles on your graph to illustrate this.

7. Suppose that $f(x)$ is an decreasing, continuous function. Write the following quantities from least to greatest:

   $$\text{LHS}_5 \qquad \text{LHS}_{10} \qquad \text{RHS}_5 \qquad \text{RHS}_{10}$$

   Explain your answer.

8. Write down the definition of the definite integral of $f(x)$ from $x = a$ to $x = b$ using a left-hand sum.

9. Explain why it is is *not true* in general that

   $$\sum_{i=1}^{n} f(a + (i-1)\Delta x)\Delta x = \sum_{i=1}^{n} f(a + i\Delta x)\Delta x$$

   but it is true that

   $$\lim_{n\to\infty} \sum_{i=1}^{n} f(a + (i-1)\Delta x)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} f(a + i\Delta x)\Delta x$$

   where $\Delta x = \frac{b-a}{n}$.

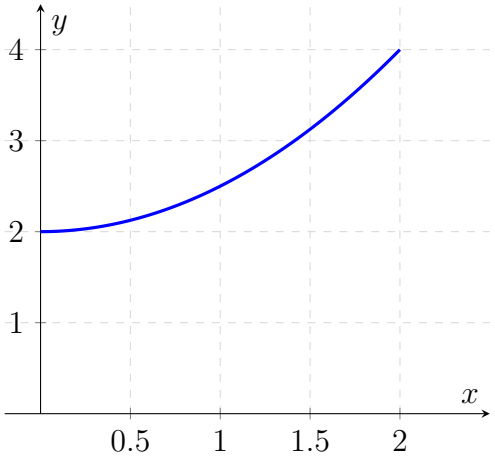
10. Consider the function $f(x) = \frac{1}{2}x^2 + 2$ on the interval $[0, 2]$.

(a) Divide the interval $[0, 2]$ into $n = 4$ subintervals of equal width. Determine the value of Δx and the endpoints of the subintervals.

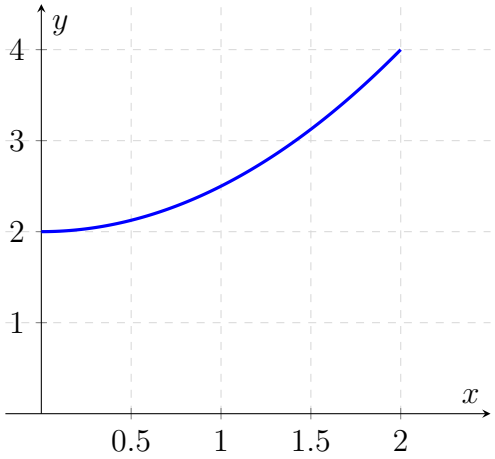
(b) On **Graph 1**, sketch and shade the rectangles used to calculate the **Left-Hand Sum** (L_4). Compute the value of LHS_4.

(c) On **Graph 2**, sketch and shade the rectangles used to calculate the **Right-Hand Sum** (R_4). Compute the value of RHS_4.

Graph 1: Left-Hand Sum

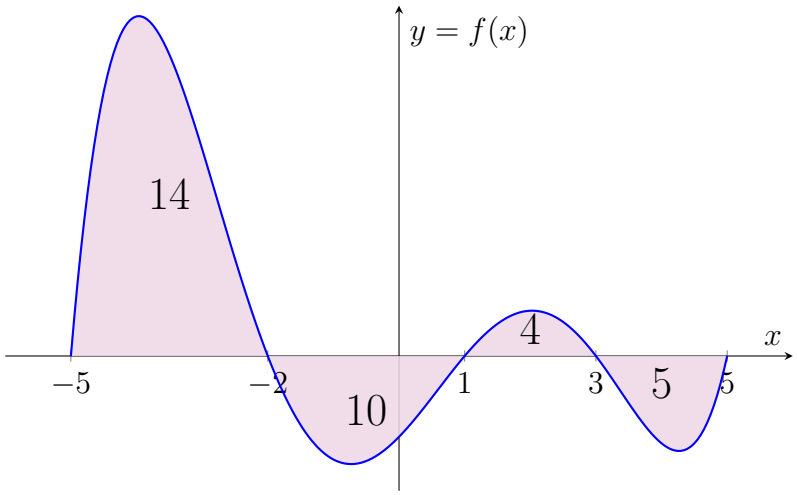


Graph 2: Right-Hand Sum



From Riemann Sums to Definite Integrals

1. The graph of $y = f(x)$ is shown below for the interval $[-5, 5]$. The areas of the four regions bounded by the curve and the x-axis are indicated.



Use the graph and the labeled areas to compute the following definite integrals:

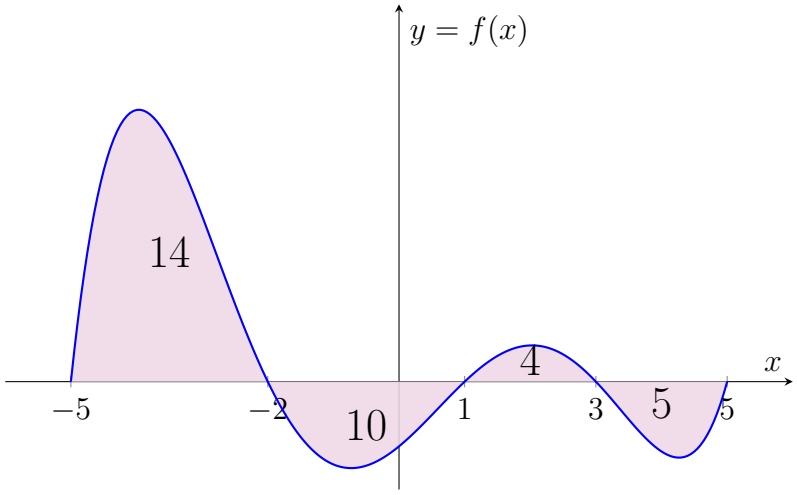
(a) $\int_{-5}^{1} f(x)\,dx$

(b) $\int_{-2}^{3} f(x)\,dx$

(c) $\int_{1}^{5} f(x)\,dx$

(d) $\int_{-5}^{5} f(x)\,dx$

2. The graph of $y = f(x)$ is shown below for the interval $[-5, 5]$. The areas of the four regions bounded by the curve and the x-axis are indicated.



Use the graph and the labeled areas to compute the following definite integrals:

(a) $\int_{-5}^{1} f(x)\,dx$

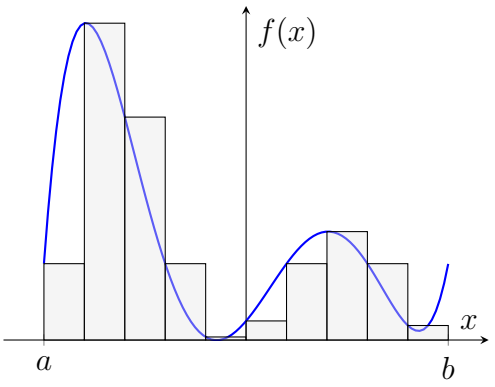
(b) $\int_{-2}^{3} f(x)\,dx$

(c) $\int_{1}^{5} f(x)\,dx$

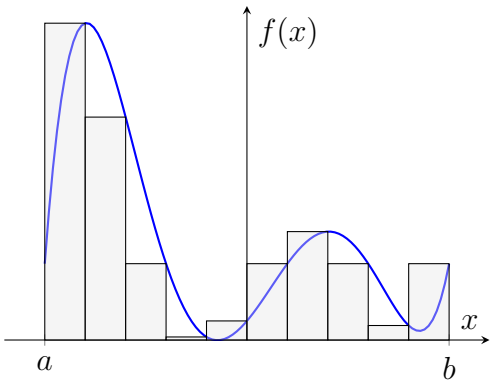
(d) $\int_{-5}^{5} f(x)\,dx$

3. Below are two copies of the graph of the same function $y = f(x)$ on the interval $[a, b]$. Each graph shows a Riemann sum approximation with $n = 10$ subintervals of equal width Δx.

Graph 1



Graph 2



(a) Identify which graph represents the **Left-Hand Sum** and which represents the **Right-Hand Sum**.

(b) On **both** graphs, locate and shade in the specific rectangle whose area is given by the term:

$$f(a + 5\Delta x)\Delta x$$

LHS/RHS Precision and Geometric Integrals

1. **Constant Function:** Compute $\int_{-1}^{4} 3\,dx$ using geometry. Sketch the graph and shade the relevant region.

2. **Linear Function:** Compute $\int_{0}^{6} (x-2)\,dx$ using geometry. Sketch the graph and shade the relevant regions.

3. **Semicircle:** Compute $\int_{-2}^{2} \sqrt{4-x^2}\,dx$ using geometry. Sketch the graph and shade the relevant region.

4. **Piecewise Mixed:** Compute $\int_{-2}^{3} f(x)\,dx$ where

$$f(x) = \begin{cases} 2x+4 & -2 \le x \le 0 \\ -2 & 0 < x \le 3 \end{cases}$$

   Sketch the graph and shade the regions.

5. **Piecewise Mixed:** Compute $\int_{-3}^{3} g(x)\,dx$ where

$$g(x) = \begin{cases} x+3 & -3 \le x \le 0 \\ -\sqrt{9-x^2} & 0 < x \le 3 \end{cases}$$

   Sketch the graph and shade the regions.

6. Let $f(x) = 2x + 7$ on the interval $[1, 6]$. Find the minimum number of equal-width subintervals n such that the difference between the Right Hand Sum and the Left Hand Sum is less than 0.01.

7. Let $f(x) = e^x$ on the interval $[0, 3]$. How many subintervals n are needed so that $|RHS_n - LHS_n| \le 0.1$? (Use $e^3 \approx 20.086$).

8. Let $f(x) = \sin(x)$ on the interval $[0, \pi/2]$. Find n such that $|RHS_n - LHS_n| < 0.05$.

9. Let $f(x) = \frac{10}{x}$ on the interval $[2, 10]$. Find n such that $|RHS_n - LHS_n| < 0.25$.

10. Let $f(x) = \sqrt{x}$ on the interval $[0, 100]$. Find n such that $|RHS_n - LHS_n| < 1$.

Midpoint and Trapezoid Sums

1. Consider $\int_0^4 2x+3\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Explain why MPS and TRAP both give exact answers here.

2. Consider $\int_0^2 x^2+1\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Does MPS overestimate this integral or underestimate it? Why?

3. Consider $\int_1^5 \frac{12}{x}\,dx$. Compute MPS_2 and TRAP_2. Sketch the rectangles for MPS_2. Does TRAP overestimate this integral or underestimate it? Why?

4. Consider $\int_0^2 10-e^x\,dx$. Compute MPS_2 and TRAP_2 to three decimal places. Sketch the rectangles for MPS_2. Does MPS overestimate or underestimate this integral? Why?

5. A student wrote the following:

   "Since $f(x)=x^2$ is increasing when $x>0$, a midpoint sum estimate of the integral $\int_2^9 x^2\,dx$ underestimates the integral."

   Is their argument correct? If so, explain why. If not, correct it.

Fundamental Theorem of Calculus

1. Which of the following concepts were involved in our proof of the Fundamental Theorem of Calculus? Circle all correct answers:

   Linear Approximation Concavity Critical Points Riemann Sums Definite Integrals Indefinite Integrals Antiderivatives

2. The Fundamental Theorem only applies to one of the following two integrals. Compute the integral it applies to, and explain why it does not apply to the other:

$$\int_1^4 \frac{1}{(x-2)^2}\,dx \qquad \int_3^4 \frac{1}{(x-2)^2}\,dx$$

3. The Fundamental Theorem applies to both the following two integrals:

$$\int_3^4 e^x\,dx \qquad \int_3^4 e^{x^2}\,dx$$

   (a) Explain why it applies to both.

   (b) Explain why, despite that, one of these can't actually be computed using the FTC.

   (c) Compute the integral the FTC applies to.

4. Explain why, while the FTC applies to the integral

$$\int_0^4 \sqrt{16-x^2}\,dx,$$

   and could be used, we don't actually need to use FTC to compute this integral. Then compute the integral exactly.

Integration

1. $\displaystyle\int \frac{\sin(\ln x)}{x}\,dx$

2. $\displaystyle\int x\cos(3x)\,dx$

3. $\displaystyle\int \sin^3 w\,dw$
   (Hint: Use the identity $\sin^2 x + \cos^2 x = 1$ to rewrite the integral before computing)

4. $\displaystyle\int_0^1 xe^{-x}\,dx$

5. $\displaystyle\int \frac{e^x}{1+e^{2x}}\,dx$

6. $\displaystyle\int_1^e \frac{(\ln x)^3}{x}\,dx$

7. $\displaystyle\int \frac{x+2}{x^2+1}\,dx$

8. $\displaystyle\int t^2 \ln t\,dt$

9. $\displaystyle\int \frac{1}{x\ln x}\,dx$

10. $\displaystyle\int_0^{\pi/4} \tan\theta\sec^2\theta\,d\theta$

11. $\displaystyle\int x^2e^x\,dx$

12. $\displaystyle\int \frac{1}{x^2+4}\,dx$

13. $\displaystyle\int \frac{\ln x}{x^2}\,dx$

14. $\displaystyle\int x^3\sqrt{x^4+1}\,dx$

15. $\displaystyle\int \sin(2x)\cos(2x)\,dx$

16. $\displaystyle\int x\ln(x+1)\,dx$

17. A student attempted to evaluate $\int xe^x \, dx$ as follows:

$u = e^x \qquad dv = xdx$
$du = e^x dx \qquad v = \frac{1}{2}x^2$

$$\int xe^x \, dx = \frac{1}{2}x^2e^x - \int \frac{1}{2}x^2e^x \, dx$$

Identify their error, then provide the correct solution.

18. A student evaluated $\int (3x+1)^4 \, dx$ as follows:

$$\int (3x+1)^4 \, dx = \frac{1}{5}(3x+1)^5 + C$$

Check: $\frac{d}{dx}(\frac{1}{5}(3x+1)^5 + C) = 3(3x+1)^4$

Identify their error, then provide the correct solution.

19. A student evaluated $\int \cos^2(x) \, dx$ as follows:

$\cos^2(x) = 1 - \sin^2(x)$

So $\int \cos^2(x) \, dx = \int 1 - \sin^2(x) \, dx = x - \frac{1}{3}\sin^3(x) + C$

Identify their error, then provide the correct solution.

20. A student evaluated $\int x\cos(x) \, dx$ as follows:

$u = x \qquad dv = \cos(x)dx$
$du = dx \qquad v = \sin(x)$

$$\int x\cos(x) \, dx = x\sin(x) + \int \sin(x) \, dx = x\sin(x) - \cos(x) + C$$

Identify their error, then provide the correct result.